



Observations of the Fermi Bubbles and the Galactic Center Excess with the Dark Matter Particle Explorer

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Abstract

The Dark Matter Particle Explorer (DAMPE) is a spaceborne high-energy particle detector that surveys the γ -ray sky above ~ 2 GeV with a peak acceptance of ~ 0.2 m² sr. With the 102 months of data collected by DAMPE, we show that the Fermi bubbles are detected at a significance of $\sim 26\sigma$ and identify a GeV excess in the direction of the Galactic center at $\sim 7\sigma$ confidence. Both spectra and morphology are consistent with those observed by the Fermi Large Area Telescope (Fermi-LAT) and the GeV excess component can be interpreted by dark matter

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annihilation with a mass of ~ 50 GeV and a velocity-averaged cross section of $\sim 10^{-26} \text{ cm}^3 \text{ s}^{-1}$ for the $\chi\chi \rightarrow b\bar{b}$ channel. Our results thus provide the first independent detection of these two intriguing diffuse γ -ray sources besides Fermi-LAT.

Unified Astronomy Thesaurus concepts: [Gamma-ray astronomy \(628\)](#); [Galactic center \(565\)](#); [Superbubbles \(1656\)](#); [Dark matter \(353\)](#)

1. Introduction

The Galactic center (GC) is the most extreme region in our Galaxy. It harbors a supermassive black hole (SMBH), dense clouds and star clusters, a wealth of supernova remnants and pulsar wind nebulae, and a large amount of dark matter (DM). The GC region attracts wide interest in the high-energy astrophysics community. It is a high-energy cosmic-ray (CR) accelerator as shown with the central point source (F. Aharonian et al. 2004; M. Chernyakova et al. 2011) and the diffuse emission within the inner $\sim 1^\circ$ region (A. Abramowski et al. 2016; V. A. Acciari et al. 2020; X. Huang et al. 2021; A. Albert et al. 2024). It is a favorable target for DM indirect searches since it is the place where the DM density peaks (G. Bertone et al. 2005; G. Bertone & D. Hooper 2018), and an extended excess is reported in the GC (E. Charles et al. 2016; S. Murgia 2020). It may also be responsible for the large-scale bubbles in the inner Galaxy, which originated from energetic activities of the SMBH in the past (M. Su et al. 2010; P. Predehl et al. 2020). In the past decades, the Fermi Large Area Telescope (Fermi-LAT) has contributed greatly to the understanding of the GC. Two particularly interesting sources are the GC excess and the Fermi bubbles.

The GC excess is an unexpected extended structure on top of the regular diffuse emission from Galactic CR sea found in Fermi-LAT data (L. Goodenough & D. Hooper 2009; V. Vitale & A. Morselli 2009; K. N. Abazajian 2011; D. Hooper & L. Goodenough 2011; D. Hooper & T. Linden 2011; K. N. Abazajian & M. Kaplinghat 2012; C. Gordon & O. Macías 2013; D. Hooper & T. R. Slatyer 2013; B. Zhou et al. 2015; F. Calore et al. 2015a, 2015b; M. Ajello et al. 2016; T. Daylan et al. 2016; X. Huang et al. 2016; M. Ackermann et al. 2017). It is an $\sim 10^\circ$ rounded excess located around the GC and has a spectrum that peaks at $\sim 1\text{--}3$ GeV with a high-energy tail. The excess is robust against systematic uncertainties in analyses of Fermi-LAT data (F. Calore et al. 2015b; B. Zhou et al. 2015; M. Ackermann et al. 2017; Y.-M. Zhong et al. 2020; I. Cholis et al. 2022; M. Pohl et al. 2022a). The GC excess is usually explained by prompt emission from DM annihilation products (e.g., L. Goodenough & D. Hooper 2009; C. Karwin et al. 2017) or curvature emission from an unresolved millisecond pulsar (MSP) population (K. N. Abazajian 2011; N. Mirabal 2013; Q. Yuan & B. Zhang 2014; T. D. Brandt & B. Kocsis 2015; Q. Yuan & K. Ioka 2015; A. Gautam et al. 2022). Extensive morphological studies have been conducted to find out the origin of the excess (R. Bartels et al. 2018; O. Macías et al. 2018, 2019; R. K. Leane & T. R. Slatyer 2019; M. Buschmann et al. 2020; M. Di Mauro 2021; S. D. McDermott et al. 2023; D. Song et al. 2024; Y.-M. Zhong & I. Cholis 2024; E. D. Ramirez et al. 2025). According to recent hydrodynamical simulations, the DM distribution of the Milky Way (MW) is found to be elongated along the disk as well (M. M. Muru et al. 2025), reducing the morphological difference between the two hypotheses.

A DM origin may be supported by the possible GeV antiproton excess (M.-Y. Cui et al. 2017; A. Cuoco et al. 2017; Y.-Z. Fan et al. 2022; C.-R. Zhu et al. 2022; K.-K. Duan et al. 2025b) identified in AMS-02 data (M. Aguilar et al. 2016, 2025). But some DM channels may be in tension with observations of the MW dwarf spheroidal satellite galaxies (dSphs; M. Ackermann et al. 2015b; A. Albert et al. 2017; S. Ando et al. 2020; A. McDaniel et al. 2024; S. Li 2026).

The Fermi bubbles, first discovered in Fermi-LAT data, consist of two large lobes, each of which is approximately 40° wide and extends to 55° above and below the GC (M. Su et al. 2010). Their spectrum is hard from 0.1 GeV to 100 GeV and begins to soften at around 100 GeV (M. Su et al. 2010; M. Ackermann et al. 2014). The two lobes share a similar spectral shape, and the interiors of the bubbles are uniform (M. Su et al. 2010; D. Hooper & T. R. Slatyer 2013; M. Ackermann et al. 2014; R.-Z. Yang et al. 2014; S. A. Narayanan & T. R. Slatyer 2017). The Fermi bubbles are believed to originate from the GC region, but their radiation mechanism is still under debate (see K. C. Sarkar 2024, for a recent review). The bubbles could have originated from past jet activity (F. Guo & W. G. Mathews 2012; H.-Y. K. Yang et al. 2022), accretion winds (K. Zubovas & S. Nayakshin 2012; G. Mou et al. 2014), or star-driven outflows (K.-S. Cheng et al. 2011; R. M. Crocker & F. Aharonian 2011).

The DArk Matter Particle Explorer (DAMPE) is a spaceborne high-energy particle detector launched on 2015 December 17, aiming to measure CRs and γ -rays in a wide energy range (G. Ambrosi et al. 2017; Q. An et al. 2019; F. Alemanno et al. 2021, 2022a, 2025a, 2025b, 2025c, 2025d). It surveys the γ -ray sky from ~ 2 GeV to 10 TeV twice a year with a peak acceptance of $\lesssim 2000 \text{ cm}^2 \text{ sr}$ (J. Chang et al. 2017; G. Ambrosi et al. 2019). Thanks to the unprecedentedly high energy resolution (Z.-L. Xu et al. 2022), DAMPE has set stringent constraints on the γ -ray line produced by DM annihilation (F. Alemanno et al. 2022b; J.-G. Cheng et al. 2023; Y.-Z. Fan et al. 2024). In this work, we report observations of the Fermi bubbles and the GC excess with DAMPE. The two sources were only detected by Fermi-LAT prior to DAMPE, and now they are, for the first time, verified by an independent telescope. Furthermore, with higher energy resolution and a lower CR background comparable to Fermi-LAT, DAMPE is able to perform such measurements with potentially smaller systematic uncertainties. The paper is structured as follows: In Section 2, the dataset, the baseline model, and the fitting algorithm are described. We first analyze the Fermi bubbles in Section 3 and then proceed to the GC excess in Section 4. For both sources, the spectrum and morphology are studied, and the systematic uncertainties are evaluated. Finally, conclusions are made in Section 5.

2. Data Analysis

2.1. DAMPE Data

A sophisticated algorithm was developed to distinguish photons from background events with a high efficiency: the CR contamination is $\lesssim 15\%$ that of the extragalactic diffuse γ -ray emission above ~ 3 GeV and is negligible above 20 GeV (Z.-L. Xu et al. 2018). Recently, the $\nu . 6 . 0 . 3$ photon dataset was produced using the updated selection algorithm, which further reduces proton and electron contamination by $\lesssim 50\%$ and $\lesssim 15\%$, respectively, at the cost of a mere $\sim 3\%$ loss in acceptance (Z.-Q. Shen et al. 2023). The new version of the dataset also benefits from recent calibration parameters (G. Ambrosi et al. 2019; W. Jiang et al. 2020; Y.-X. Cui et al. 2023). In this work, we use this updated dataset.²⁶

We select photon events collected from 2016 January 1 to 2024 June 30 (DAMPE mission elapsed time of 94,608,001–362,793,602 s). To avoid pileup of the instrument, the observations collected within the SAA or during strong solar flares are excluded. We only choose high-energy trigger events above 2 GeV. After the data reduction, around 359,000 photon events remain.

Instead of fitting the two targets based on the all-sky data, which can greatly suffer from systematic uncertainties, we define different regions of interest (ROIs) for them: the central $120^\circ \times 120^\circ$ region excluding 10° of the Galactic plane for the Fermi bubbles, whereas a $40^\circ \times 40^\circ$ one excluding 2° of the Galactic plane for the GC excess. These ROIs enclose the main parts of the sources while not introducing too much astrophysical background emission. Intensity maps for the two ROIs are presented in Sections 3 and 4. We further partitioned the data into spatial and energy bins to perform binned likelihood analyses. For the GC excess, the photons from 2 to 200 GeV are binned into 400×400 pixels in the plate carrée (CAR) projection (0.10° pixel size) and 20 logarithmically spaced energy bins. For the Fermi bubbles, HEALPix pixelization²⁷ (K. M. Górski et al. 2005) with an order of 7 ($\approx 0.46^\circ$ pixel size) is adopted to facilitate the convolution of instrumental responses. The energy range also extends up to 500 GeV to better fit the high energy of the bubbles, and the data are split into 24 energy bins.

To reduce the number of free parameters, the bright sources listed in the DAMPE point-source catalog (DAMPE Collaboration 2026, in preparation) are masked. In the analysis of the Fermi bubbles, circular masks with radii of 1.5° are applied, which can remove 95% of the point-source emission at 2 GeV (K.-K. Duan et al. 2025a). On the other hand, masks that are dependent on the source significance are employed for the GC excess to make better use of the observed data: 0.7° (0.3°) is chosen for those with a test statistic (TS) larger (smaller) than 49 in the catalog. The data selection and binning are summarized in Table 1.

2.2. γ -Ray-emitting Components

Several components make up the GeV γ -ray sky observed by DAMPE, including the Galactic diffuse emission (GDE), the isotropic background, and many pointlike, extended, and

Table 1

Summary Table of the Data Selection for the Analyses of the Fermi Bubbles (Second Column) and the Galactic Center Excess (Third Column)

Selection	Fermi Bubbles	GC Excess
ROI	$5^\circ \leq b \leq 60^\circ$, $ l \leq 60^\circ$	$1^\circ \leq b \leq 20^\circ$, $ l \leq 20^\circ$
Spatial bins	HEALPix, order 7	CAR, 400×400
Energy bins	2–500 GeV, 24 bins	2–200 GeV, 20 bins
Point-source mask	1.5° (TS ≥ 25)	0.3° (25 $\leq$ TS < 49), 0.7° (TS ≥ 49)

Note. The observed intensity maps in the ROIs of the Fermi bubbles and GC excess in the CAR projection are shown in Figures 1 and 6, respectively. The point-source masks are only applied around the sources listed in the DAMPE 8.7 yr catalog. Details can be found in Section 2.1.

diffuse sources. In this subsection, we will describe the baseline model, summarized in Table 2, and leave the systematic uncertainties for the following sections.

GDE originates from the interactions of Galactic CRs with interstellar gas via inelastic nucleon–nucleon collisions and electron bremsstrahlung, and with the ISRF via IC processes (F. Acero et al. 2016b). In the baseline model, we employ the GDE emission model calculated with the CR propagation code Galprop²⁸ (A. W. Strong & I. V. Moskalenko 1998; A. W. Strong et al. 2000; T. A. Porter et al. 2008) using the parameter set named as Lorimer_z10_Ts150_P7 (denoted as $^{\text{S}}\text{L}^{\text{Z}}10^{\text{T}}150$ hereafter) developed in the first Fermi-LAT supernova remnant catalog (ISC; F. Acero et al. 2016a).²⁹ The model is constructed under the assumptions of the CR source distributions following that traced by pulsars (D. R. Lorimer et al. 2006), a propagation halo with a height of $z_h = 10$ kpc and a radius of $R_h = 30$ kpc, and a uniform H I spin temperature of $T_s = 150$ K. The hadronic component and bremsstrahlung component, spatially correlated with the gas density, are combined into gas-correlated templates associated with either neutral hydrogen H I or molecular hydrogen H₂.³⁰ To be more flexible in the analyses of the bubbles, the H I map is further split into three annuli: an inner ring spanning Galactocentric radii of 0–8 kpc, a local ring spanning 8–10 kpc, and an outer ring spanning 10–30 kpc. The IC template is made by simply summing up the contributions of all the ISRF components.

Loop I is a giant structure spanning $\sim 100^\circ$ on the sky visible in radio (E. M. Berkhuijsen 1971; P. A. R. Ade et al. 2016), X-ray (R. J. Egger & B. Aschenbach 1995; P. Predehl et al. 2020) and γ -ray bands (J.-M. Casandjian & I. Grenier 2009). It is originally proposed to be associated with the Sco–Cen OB association at a distance of 120–140 pc (C. J. Salter 1983), but growing evidence shows that Loop I may be located beyond the OB association (P. A. R. Ade et al. 2016) and might even have a GC origin (Y. Sofue 2015). Irrespective of its origin, we adopt the geometric model from M. Wolleben (2007) following the previous works (M. Ackermann et al. 2014, 2017; M. Ajello et al. 2016). This model is made up of two spherical shells, each of which is described with the coordinates (ℓ_s , b_s) of the shell center, the distance to the

²⁶ The photon data above 3 GeV up to 2024 January 1 are publicly available at <https://dampe.nssdc.ac.cn>. The filters of the South Atlantic Anomaly (SAA) and solar flares have been applied in the public dataset in default.

²⁷ <http://healpix.sourceforge.net>

²⁸ <https://galprop.stanford.edu/>

²⁹ https://fermi.gsfc.nasa.gov/ssc/data/access/lat/1st_SNR_catalog/

³⁰ Ionized hydrogen and dark gas are incorporated in the H I map (M. Ackermann et al. 2012; F. Acero et al. 2016a). Helium and heavier element gases are assumed to be mixed with hydrogen uniformly (J.-M. Casandjian 2015).

Table 2

Summary Table of the Template Maps Adopted in the Data Analyses of the Fermi Bubbles (Second Column) and the Galactic Center Excess (Third Column)

Component	Analysis of Fermi Bubbles (Figure 15)	Analysis of GC Excess (Figure 16)
H I + H II Gas	Galprop, binned into three rings: 0–8 kpc (free), 8–10 kpc (free), and 10–30 kpc (fixed)	Galprop, all rings combined: 0–30 kpc (free)
H ₂ Gas	Galprop, all rings combined (free)	Galprop, all rings combined (free)
Inverse Compton (IC)	Galprop, all interstellar radiation field (ISRF) combined (free)	Galprop, all ISRF combined (free)
Loop I	Geometric template (free)	Geometric template (free)
Isotropic	Proportional to exposure map (free)	Proportional to exposure map (free)
Point Sources	DAMPE point sources (fixed)	DAMPE point sources (fit and fixed) and weak Fermi-LAT sources (free)
Fermi Bubbles	Flat template in this work (free)	Flat template (free with priors)
GC Excess	...	DM annihilation template (free)

Note. The baseline Galprop templates are labeled $^{\text{S}}\text{L}^{\text{Z}}10^{\text{T}}150$, which are from F. Acero et al. (2016a). The maps of some components and the mask in the analyses of Fermi bubbles and GC excess are shown in Figures 15 and 16, respectively, in the Appendix. More descriptions on the components can be found in Section 2.2.

center d_s , and the inner and outer radius (r_{in} , r_{out}). The parameter sets (ℓ_s , b_s , d_s , r_{in} , r_{out}) for the two shells are (341° , 3° , 78 pc, 62 pc, 81 pc) and (332° , 37° , 95 pc, 58 pc, 82 pc). The γ -ray emissivities of the shells are assumed to be the same. We make the Loop I spatial template by integrating the emission from the shells along the line of sight.

The isotropic diffuse background comprises unresolved extragalactic sources, extragalactic diffuse emission, and misclassified CR particles (M. Ackermann et al. 2015a; M. Ajello et al. 2015). DAMPE has a lower CR contamination but a worse point-source sensitivity than Fermi-LAT, so the spectrum of the isotropic background is expected to be different from that of Fermi (M. Ackermann et al. 2015a). Here, we adopt a PL spectral shape in the global fitting.

We also consider point sources within the ROIs. Around 360 point sources are detected with TS values larger than 25 in 8.7 yr of DAMPE photon data, given the standard Fermi-LAT Galactic interstellar emission model (F. Acero et al. 2016b). For the analysis of Fermi bubbles, we compute an expected counts cube based on the fitted spectral parameters listed in the source catalog. Since the masks are large enough to remove the majority of emissions from the point sources, we keep the model fixed in the fitting. However, for the analysis of GC excess, the masks are not that large and their spectral parameters may be relevant. To ensure the parameters are compatible with the background emission, we first fit the point sources without masking them, and then adopt the fitted parameters in the analysis afterward when the point-source masks are applied. Weak sources below the DAMPE sensitivity are also accounted for when analyzing the GC excess. We create a new template containing the sources listed in the Fermi-LAT 14 yr catalog³¹ (4FGL-DR4; S. Abdollahi et al. 2022; J. Ballet et al. 2023) but not in the DAMPE catalog. We use a power law (PL) as a scale and fit the component as a whole in the analysis.

The last two components are the Fermi bubbles and GC excess, which are the targets in this work. Since the two components do not have counterparts at wavelengths other than γ -rays, we derive them from the DAMPE data. First, we analyze the Fermi bubbles and extract their spatial templates from the residual map. Then, based on the obtained template of the bubbles, we conduct the GC excess analysis, where the

spatial models of DM annihilation and MSPs are tested. More details on the construction of the templates will be presented in the following sections.

2.3. Binned Likelihood Analysis

The total γ -ray intensity I_{tot} in direction (ℓ, b) at energy E is the linear combinations of the components:

$$I_{\text{tot}}(\ell, b, E; \Theta) = \sum_m f_m(E; \Theta_m) I_m(\ell, b, E), \quad (1)$$

where $f_m(E; \Theta_m)$ is the scaling factor parameterized with Θ_m for the template of the m th component $I_m(\ell, b, E)$. In the global fitting, we use a PL with an exponential cutoff (PLEC) spectrum for the bubbles and a log-parabola spectrum for the GC excess. For other components, PL scaling factors are adopted.³²

We utilize the DAMPE γ -ray analysis toolkit DmpST (K.-K. Duan et al. 2019) to apply the instrumental response functions (IRFs). The recent version of this software implements up-to-date IRF calibration parameters (Z.-Q. Shen et al. 2024; K.-K. Duan et al. 2025a). We calculate the exposure map $\varepsilon(\ell, b, E)$ considering the azimuth-angle dependence of the effective area. Then the total intensity map is multiplied by the exposure map, convolved with the point-spread function (PSF), and integrated to achieve the expected counts map in the energy bin $\tilde{\mu}_{\text{tot}} = \int dE [I_{\text{tot}} \cdot \varepsilon] \cdot \text{PSF} \Delta\Omega$, where $\Delta\Omega$ is the solid angle of the pixel.

Binned likelihood analysis is performed in this work. The likelihood function $\mathcal{L}(\Theta)$ is defined as (J. R. Mattox et al. 1996; K.-K. Duan et al. 2019)

$$\ln \mathcal{L}(\Theta) = \sum_{ij} [N_{ij} \ln(\tilde{\mu}_{\text{tot},ij}) - \tilde{\mu}_{\text{tot},ij}], \quad (2)$$

where N_{ij} and $\tilde{\mu}_{\text{tot},ij} \equiv \tilde{\mu}_{\text{tot}}(\hat{s}_i, E_j; \Theta)$ are the observed and expected photon counts in the i th pixel centered at coordinate $\hat{s}_i$ and the j th energy bin, respectively. The likelihood is optimized with the Minuit algorithm (F. James & M. Roos 1975).

³¹ https://fermi.gsfc.nasa.gov/ssc/data/access/lat/14yr_catalog/gll_psc_v32.xml

³² See https://fermi.gsfc.nasa.gov/ssc/data/analysis/scitools/source_models.html for the definitions of the spectral models.

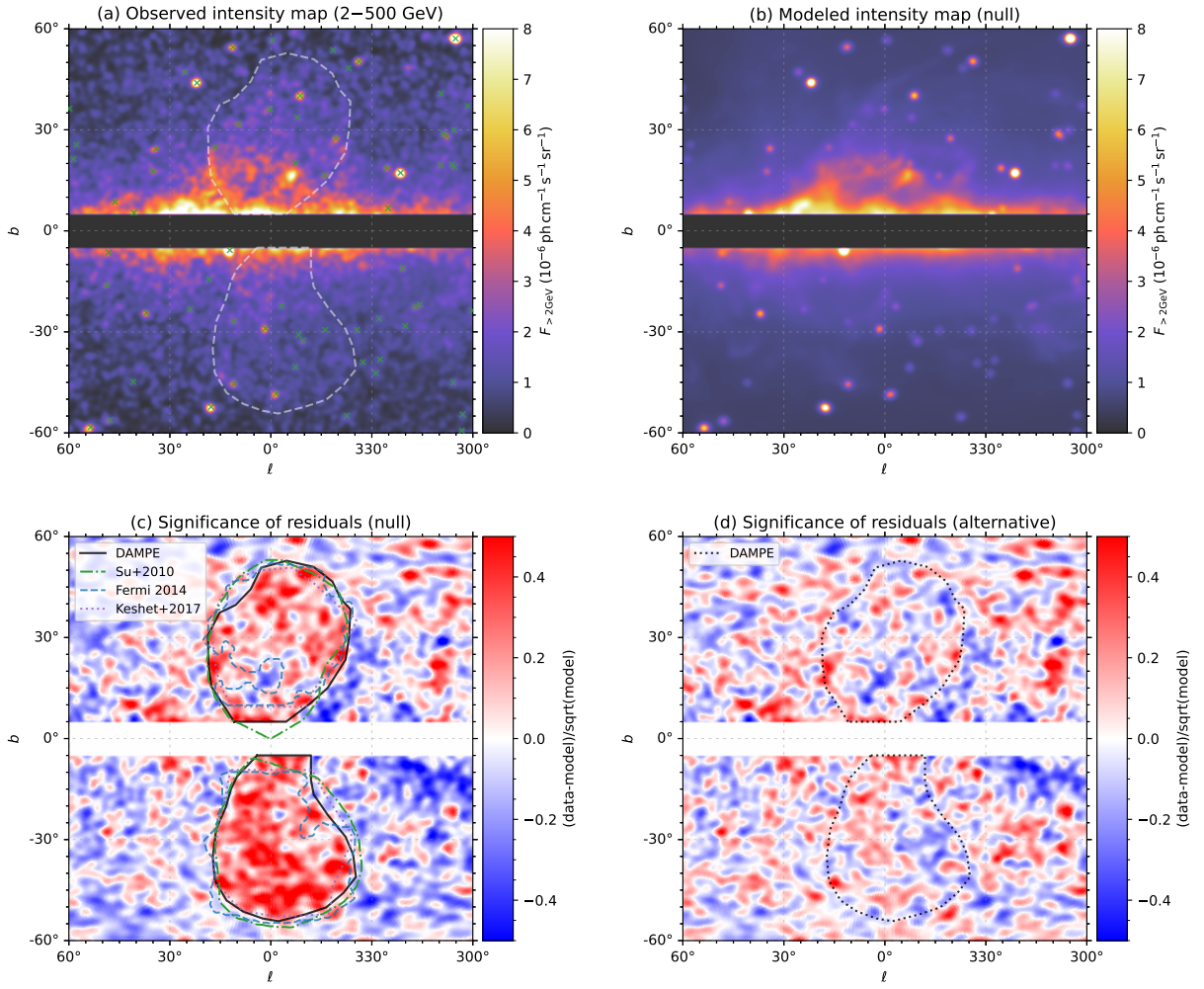


Figure 1. Upper panels: intensity map from 2 to 500 GeV in the CAR projection, smoothed with a Gaussian kernel with $\sigma = 0.75^\circ$ for (a) the observed data and (b) best-fit model without the bubbles. The Galactic plane region ($|b| \leq 5^\circ$) is masked. The green crosses mark the point sources in the DAMPE catalog, whereas the gray contour encloses the Fermi bubbles. Lower panels: the significance of the residual maps smoothed with a 1° Gaussian kernel for the models (c) without and (d) with the bubbles. The black curve shows the boundary derived in this work, while the other lines represent the ones from Fermi-LAT.

3. Fermi Bubbles

3.1. Defining a Template for the Bubbles

The photon events are selected from the $120^\circ \times 120^\circ$ ROI defined in Section 2.1. In Figure 1(a), we present the observed flux map integrated above 2 GeV, which is simply the counts map divided by the exposure time, i.e., $F(\hat{s}_i, E > 2 \text{ GeV}) = \int dE F(\hat{s}_i, E) \approx \sum_j N_{ij} / [\varepsilon(\hat{s}_i, E_j) \Delta\Omega_i]$. To remove the background components, we set the baseline model without the bubbles as the null hypothesis and perform the binned likelihood analysis. The fitted components are labeled in Table 2. Both the Galactic plane and point sources are masked. Figure 1(b) shows the flux map given the optimized null model (see Figure 17 in the Appendix for the mean flux of the components in the global fittings). Most of the emission is well described by the model, as we expected. However, if the null model is subtracted from the data, some weak excess is visible in the region of the Fermi bubbles. Figure 1(c) shows the significance of the residuals $(N_i - \tilde{\mu}_i) / \sqrt{\tilde{\mu}_i}$, where N_i and $\tilde{\mu}_i$ are the observed and predicted photon counts in the i th pixel integrated over 2–500 GeV, respectively. The excess is in good agreement with the boundaries extracted from the Fermi-LAT data, including the green dotted-dashed line from

M. Su et al. (2010), the blue dashed line from M. Ackermann et al. (2014), and the purple dotted line from U. Keshet & I. Gurwich (2017). Besides, the normalizations of the isotropic and IC components are exaggerated in the fitting, which compensates for the absence of the bubbles and leads to a photon deficit outside the boundary.

Considering the slight difference between the residuals and the previous templates, we construct a new template based on the DAMPE data. Instead of deriving the template directly from the residuals of the null model, whose backgrounds may be biased due to not including the Fermi bubbles, we adopt the one from M. Su et al. (2010) as the initial template, fit the data, and use the model subtracting the bubbles to calculate the residuals. The residuals in significance are calculated using the observed and fitted (without bubbles) counts map integrated over energies = 2–500 GeV. We make a distribution of the values inside and outside the bubble region, denoted as the bubble region and background region, respectively, as shown in Figure 2 (left). To determine the threshold for the template of the bubbles, the background histogram is fitted using a Gaussian profile. We only fit negative fluctuations in the histogram and assume the mean value of the Gaussian to be zero. The best-fit Gaussian profile, whose

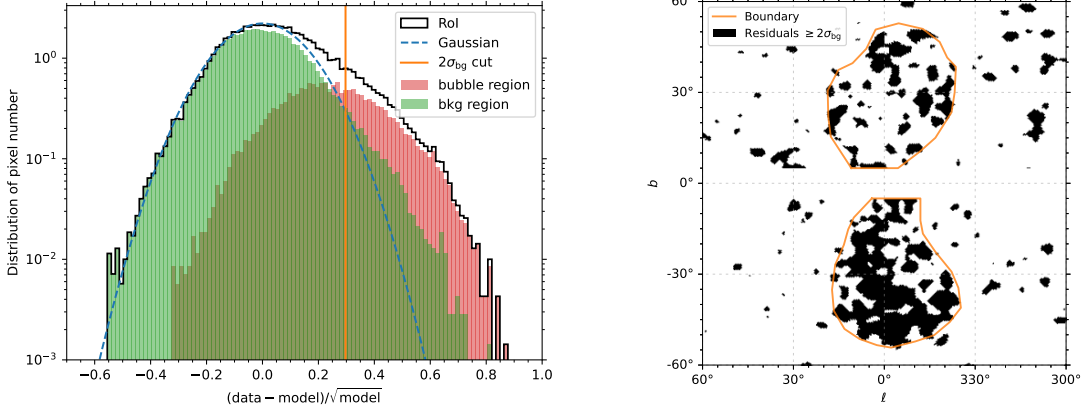


Figure 2. Left panel: distribution of the values in the significance map. The map, in HEALPix projection, is smoothed with a 1° Gaussian kernel. The black histogram shows the distribution of pixel numbers within the ROI, whereas the red and green ones show those within and outside the bubbles, respectively. The blue dashed line is a Gaussian profile with a mean value of zero fitted to the background histogram. The orange solid line represents the $2\sigma_{bg}$ cut adopted to define the bubbles' boundary. Right panel: map with significance larger than the $2\sigma_{bg}$ cut. The bubble template in this work consists of two polygons whose boundaries are illustrated with the solid orange lines.

width is $\sigma_{bg} = 0.15$, is shown with a blue dashed line. We make a map with significance exceeding $2\sigma_{bg}$ as given in Figure 2 (right). The southern bubble is clearly traced in the map, but the northern one is less obvious because of the strong background emission in the northern hemisphere. Unlike M. Ackermann et al. (2014), who directly define their template with a significance map above a threshold, we define the bubbles as two simply connected polygons because of the relatively low statistics of DAMPE data. Two criteria are used to determine the boundaries: they contain most of the significant excess in the region of the bubbles, and, at the same time, they do not greatly deviate from the template of M. Su et al. (2010). What we achieve is a flat template enclosed by the boundaries as illustrated with the orange solid line in Figure 2 (right) and with the black line in Figure 1. The derived lobes closely resemble the previous results (with a slight difference of $\sim 6\%$).

We further inspect the template of the Fermi bubbles. First, we check the residual map given the alternative model, including the new template. As shown in Figure 1(d), the residual map is almost consistent with the statistical fluctuations, confirming the good fit of the flat template to the data. A mild excess can be spotted in the eastern part of the southern bubble and is likely due to the cocoon discovered in the Fermi-LAT data (M. Su & D. P. Finkbeiner 2012). Then, we derive the flux near the boundary of the bubbles. We subtract the best-fit alternative model from the data, add the bubble model back to the residuals, and calculate the average flux as a function of angular distance ϕ from the edge using the residual map. The average flux is calculated using $F_{res, > 2\text{GeV}} = \sum_j [\sum_{i \in \text{Ring}_k} N_{res,ij}] / [\sum_{i \in \text{Ring}_k} \varepsilon_{ij} \Delta\Omega_i]$, where $N_{res,ij}$ is the residual counts in the i th pixel and j th energy bin, and Ring_k represents the k th angular distance ring. The residual flux in the high-latitude region with $|b| > 20^\circ$ is drawn with the blue points in Figure 3. The edge of the bubbles is sharp. We find the flux evolution around the edge consistent with a template with a sharp edge convolved with the PSF (green dashed line). Figure 3 also shows flat residual flux inside the bubbles, which is well consistent with the average flux for the bubbles as a whole (orange band). To conclude, the sharp edge and the uniform flux support the flat bubbles template.

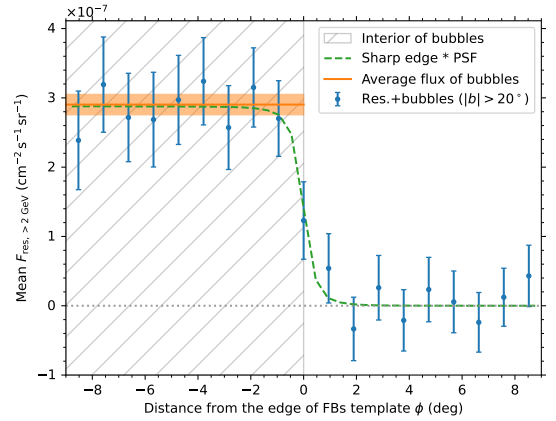


Figure 3. The average residual flux as a function of angular distance to the edge in the high-latitude region with $|b| > 20^\circ$ (blue points). The residuals are obtained by extracting the best-fit model except for the bubbles from the data. The negative (positive) x -axis values represent the region inside (outside) the bubbles. The green dashed line is the sharp edge convolved with the PSF. The orange band is the average flux of the bubble template as a whole.

This template is available at ScienceDB at doi:[10.57760/sciencedb.space.03534](https://doi.org/10.57760/sciencedb.space.03534).

3.2. Spectral Analysis

Once we have the new template of the bubbles, we can perform the bin-by-bin analyses to derive the spectral energy distribution (SED). To increase the statistics, several energy bins are combined together in fittings: two (three) bins below (above) 126 GeV. In each combined energy bin, the spectral indices of the sources are kept as those from the global fit, while the normalizations are fitted. The average fluxes of the components within the ROI are given in the left panel of Figure 4. The bubbles only make up 3%–10% of the γ -ray photons in the ROI, but they can be well distinguished from other components due to their unique morphology. The right panel of Figure 4 presents the SED values and the statistical uncertainties of the bubbles with red points. They are close to those from Fermi-LAT (M. Ackermann et al. 2014) as drawn with blue points, exhibiting a hard spectrum with a cutoff at ~ 100 GeV. The TS of the source is $\text{TS} \equiv -2\sum_i \ln(\hat{\mathcal{L}}_{\text{null},i} / \hat{\mathcal{L}}_{\text{alt},i}) = 757.4$, where $\hat{\mathcal{L}}_{\text{null},i}$ and $\hat{\mathcal{L}}_{\text{alt},i}$ are the

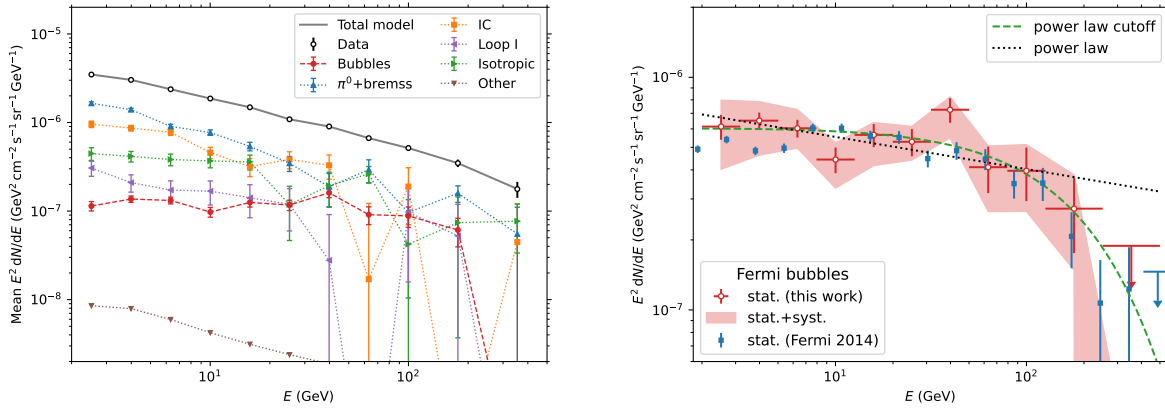


Figure 4. Left panel: the average fluxes of the observed data (black hollow points) and various fitted components in the baseline model (solid points connected with lines) within the ROI. The red points show the flux of the bubbles. The masked Galactic plane and point sources are excluded from the calculation. Right panel: the SED of the Fermi bubbles. The red points show the best-fit SED from DAMPE given the baseline model, whereas the blue points show those from Fermi-LAT (M. Ackermann et al. 2014). A 95% upper limit is given when the TS value is below nine. The red band exhibits the total errors from the statistical and systematic uncertainties. The optimal single PL model and PL cutoff model are also presented with black dotted and green dashed lines, respectively.

likelihood values for the optimal null and alternative models in the i th SED bin, respectively. Since the TS values follows the χ^2 distribution with 11 degrees of freedom in the null hypothesis (S. S. Wilks 1938), the significance of the bubbles is 26.5σ .

We estimate the systematic uncertainty of the spectrum induced by inaccurate background models. First, the GDE model is tested. The Galprop GDE model is derived assuming that the CR particles are injected following a CR source distribution, propagate throughout the Galaxy within a cylindrical propagation halo, and interact with gas and photons to produce γ -rays. Two CR source distribution tracers are tested, either from supernova remnants (G. L. Case & D. Bhattacharya 1998) or from pulsars (D. R. Lorimer et al. 2006). The height of the propagation halo ($z_h = 4$ kpc or $z_h = 10$ kpc) and the opacity of the H I gas (opaque with $T_s = 150$ K spin temperature or optically thin) are also verified. Second, we substitute the geometric Loop I template with a map of 408 MHz radio continuum around the GC (C. G. T. Haslam et al. 1982; M. Remazeilles et al. 2015).³³ The map is further restricted to the region of $|\ell| < 70^\circ$ and $5^\circ < |b| < 90^\circ$ where Loop I is prominent. Furthermore, the bubble template is also tested. The bubble templates from M. Su et al. (2010) and M. Ackermann et al. (2014) are taken as alternatives. We construct $8 \times 2 \times 3 - 1 = 47$ alternative models in total by changing the three factors of uncertainties one by one and repeating the bin-by-bin analyses to construct the SEDs. We find the TS value of the bubbles ranges from 521 to 974 in these models. The systematic uncertainty of the spectrum is defined with the envelope of all the tested models, which is listed in Table 3. The red band in Figure 4 (right) shows the root sum square of the statistical and systematic errors.

A curved spectrum is suggested in the SED. We fit the likelihood profiles in these energy bins with the PLEC model: $dN/dE \propto E^{\Gamma_1} \exp(-E/E_{\text{cut}})$. The best-fit spectrum for the baseline model is shown with the green dashed line in Figure 4 (right). Compared with the single PL model, the TS value difference is $\text{TS}_{\text{curv}} \equiv \text{TS}_{\text{plec}} - \text{TS}_{\text{pl}} = 9.3$, showing the spectrum is curved at 3σ significance (S. Abdollahi et al. 2022).

Table 3

The Differential Spectrum per Unit Angle for the Fermi Bubbles, as in the Right Panel of Figure 4

E Range (GeV)	$E^2 dN/dE$ ($10^{-7} \text{ GeV cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$)	TS Value
2.00–3.17	$6.14 \pm 0.75^{+1.68}_{-1.98}$	72.6
3.17–5.02	$6.51 \pm 0.54^{+1.23}_{-1.80}$	160.4
5.02–7.95	$6.04 \pm 0.53^{+1.11}_{-0.94}$	147.5
7.95–12.6	$4.43 \pm 0.56^{+0.56}_{-0.94}$	69.9
12.6–20.0	$5.65 \pm 0.64^{+0.36}_{-1.34}$	90.2
20.0–31.6	$5.28 \pm 0.71^{+0.50}_{-0.52}$	68.6
31.6–50.1	$7.25 \pm 0.88^{+0.45}_{-1.54}$	92.6
50.1–79.4	$4.11 \pm 0.93^{+0.23}_{-1.13}$	25.2
79.4–126	$3.96 \pm 1.03^{+0.56}_{-0.82}$	19.4
126–251	$2.72 \pm 0.97^{+0.49}_{-0.78}$	11.0
251–500	< 1.89	0.0

Note. The first and second errors are statistical and systematic uncertainties, respectively. The third column is the TS values in the energy window given the baseline background model.

Taken the systematic uncertainty into account, the optimal spectral index is $\Gamma_1 = -1.99 \pm 0.06[\text{stat.}]^{+0.10}_{-0.09}[\text{syst.}]$, the cutoff energy is $E_{\text{cut}} = 204^{+120}_{-60}[\text{stat.}]^{+63}_{-48}[\text{syst.}]$ GeV, and the integrated intensity above 2 GeV is $(2.91 \pm 0.15[\text{stat.}]^{+0.58}_{-0.69}[\text{syst.}]) \times 10^{-7} \text{ ph cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$. So the luminosity of the bubbles with $|b| > 5^\circ$ is $\approx (3.15 \pm 0.17[\text{stat.}]^{+0.38}_{-0.56}[\text{syst.}]) \times 10^{37} \text{ erg s}^{-1}$ above 2 GeV, assuming the distance to the center of each lobe ($|b| = 25^\circ$) is $R \approx 9.14$ kpc (M. Ackermann et al. 2014) and the solid angle is $\Omega \approx 0.79$ sr.

We further split the bubbles into two templates and assign the free parameters for each of the lobes. A similar likelihood analysis is performed, and the derived SEDs are shown in Figure 5. The points are the spectra of the lobes given the baseline background model, and the color bands represent the total uncertainty. The TS values for the northern and southern lobes are 236.8 and 533.5, respectively, corresponding to 14.0σ and 22.0σ significance. The northern lobe is less significant than the other one because the background emission in the northern sky is stronger (Figure 1(b)). We fit the SEDs with the PLEC model. The spectrum of the northern lobe can be described with an index of $\Gamma_1 = -1.95 \pm 0.12[\text{stat.}]$ and

³³ https://lambda.gsfc.nasa.gov/data/foregrounds/haslam_2014/haslam408_dsds_Remazeilles2014.fits

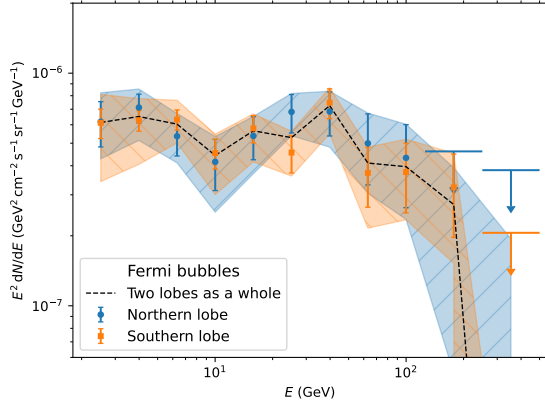


Figure 5. The SEDs of the northern (blue) and southern (orange) lobes. The points show the baseline spectrum and the statistical uncertainties, whereas the bands give the total errors. The black dashed line is the best-fit SED of the whole bubbles.

$E_{\text{cut}} = 174^{+220}_{-70}$ [stat.] GeV, whereas the southern lobe has a spectrum with $\Gamma_1 = -2.01 \pm 0.08$ [stat.] and $E_{\text{cut}} = 221^{+190}_{-80}$ [stat.] GeV. The spectra of the two lobes are in excellent agreement, supporting the identical physical origin of the two lobes.

4. Galactic Center Excess

4.1. Galactic Center Excess in the Baseline Model

To search for emission from the GC excess, we use a smaller ROI, 40° wide as defined in Section 2.1, than that for the Fermi bubbles. Figure 6(a) shows the flux map, which is corrected for exposure time and integrated above 2 GeV. The point sources in the catalog are marked as green crosses. We first fit the data globally with the baseline null model, which contains the gas templates, IC template, Loop I, the bubbles, point sources, weak point sources, and isotropic emission. Notably, the baseline Fermi bubbles template only has a high-latitude part, but we will later incorporate a lower-latitude part to evaluate the impact on GC excess. A description of the baseline model can be found in Section 2.2. In the fitting, a mask for the Galactic plane ($|b| \geq 1^\circ$) and sources is adopted to reduce the systematic uncertainties on the diffuse emission and point sources. To reduce correlations between parameters, we use the values and errors of the Fermi bubbles in Section 3 as the priors. The normalizations of the H I and H₂ are assumed to differ by one single free factor throughout the energy range to account for the CO-to-H₂ conversion factor X_{CO} . The fitted components are labeled in Table 2. Figure 6(b) shows an integrated flux map of the best-fit null model (see Figure 18 in the Appendix for the mean component flux in the global fittings). Figure 6(c) presents the significance of the residual given the null model. An extended excess is visible within the innermost 5° region.

To test the alternative hypothesis, we model the excess by DM annihilation with a template proportional to the J -factor map:

$$J(\ell, b) \equiv \int_{\text{l.o.s.}} \rho_{\text{dm}}^2(r(\ell, b, s)) ds, \quad (3)$$

where s is the line-of-sight (“l.o.s.”) distance and r is the Galactocentric distance. The J -factor map traces the prompt emission emitted by annihilating DM particles. We assume the

DM follows the generalized Navarro–Frenk–White (gNFW) density profile (J. F. Navarro et al. 1996):

$$\rho_{\text{dm}}(r) = \frac{\rho_0}{(r/r_s)^\gamma (1 + r/r_s)^{3-\gamma}}. \quad (4)$$

To be compatible with previous works, we set a scale radius of $r_s = 20$ kpc and an inner density slope of $\gamma = 1.2$. The normalization ρ_0 is determined by $\rho_{\text{dm}}(R_0) = 0.4 \text{ GeV cm}^{-3}$ (R. Catena & P. Ullio 2010) and $R_0 = 8.5$ kpc (A. M. Ghez et al. 2008). There are indeed updated measurements on the local DM density (e.g., P. J. McMillan 2017; P. F. de Salas & A. Widmark 2021) and distance to the GC (T. Do et al. 2019; K. Abd El Dayem et al. 2024). If we take the parameters ($R_0, r_s, \rho_{\text{dm}}(R_0)$) = (8.28 kpc, 25 kpc, 0.38 GeV cm^{-3}) (P. J. McMillan 2017; K. Abd El Dayem et al. 2024), the J -factor would be decreased by $\sim 22\%$ on average. But the change can be mostly absorbed into the normalization of the GC excess in the fittings.

We add the GC excess to the model and fit the model to the observed data. Figure 6(d) shows a residual map using the alternative model. The residuals in the central 5° region are well consistent with the statistical fluctuations. To further visualize the spatial distribution of the excess, we split the residual map according to the angular distance to the GC and calculate the average flux within several annuli of width 1° . In Figure 7, the gray points represent the average flux of the residuals for the model including the GC excess, which confirms that the residuals are mostly flat. The weak negative residuals in the inner region suggest the intensity of the excess is slightly overstated due to the complex background. We also readd the GC excess model to the residual map and show the average flux with blue points. The excess is indeed peaked at the GC, and the radial distribution is consistent with the baseline J -factor map.

Bin-by-bin analysis is conducted to derive the SED of GC excess. Several energy bins are combined together to increase the statistics: two bins below 20 GeV, three bins between 20 and 80 GeV, and four bins above 80 GeV. Based on the global-fit model, we fix the spectral indices, tie the normalizations of H I and H₂ together while keeping the ratio, and fit the parameters in each combined energy bin. The left panel of Figure 8 shows the mean flux of the γ -ray components in the ROI. On average, the GC excess contributes more photons than the Fermi bubbles and point sources below ~ 8 GeV. But it is still much weaker than the GDE and therefore is prone to the systematics of the background model. The SED of the GC excess at 5° from GC, given the baseline background model, is shown in the right panel of Figure 8 and listed in Table 4. The DAMPE results are in good agreement with the spectra extracted from Fermi-LAT data (F. Calore et al. 2015b; B. Zhou et al. 2015; I. Cholis et al. 2022).³⁴ A slight tension might present in the energy range of 8–20 GeV since the upper limits are below the observed flux from Fermi-LAT. The TS value for the GC excess in the whole energy range is 80.1, showing 7.5σ significance given eight degrees of freedom (S. S. Wilks 1938). A weak excess with a TS value of 6.3 is presented between 20 and 80 GeV, which might suggest a

³⁴ F. Calore et al. (2015b) and I. Cholis et al. (2022) present spectra of the GC excess averaged over the ROI of $|\ell| \leq 20^\circ$ and $2^\circ \leq |b| \leq 20^\circ$, so 4.41 is multiplied to their SEDs to convert to the ones measured at 5° from the GC. B. Zhou et al. (2015) show the mean SED within 10° of the GC, hence a conversion factor of 0.766 is adopted.

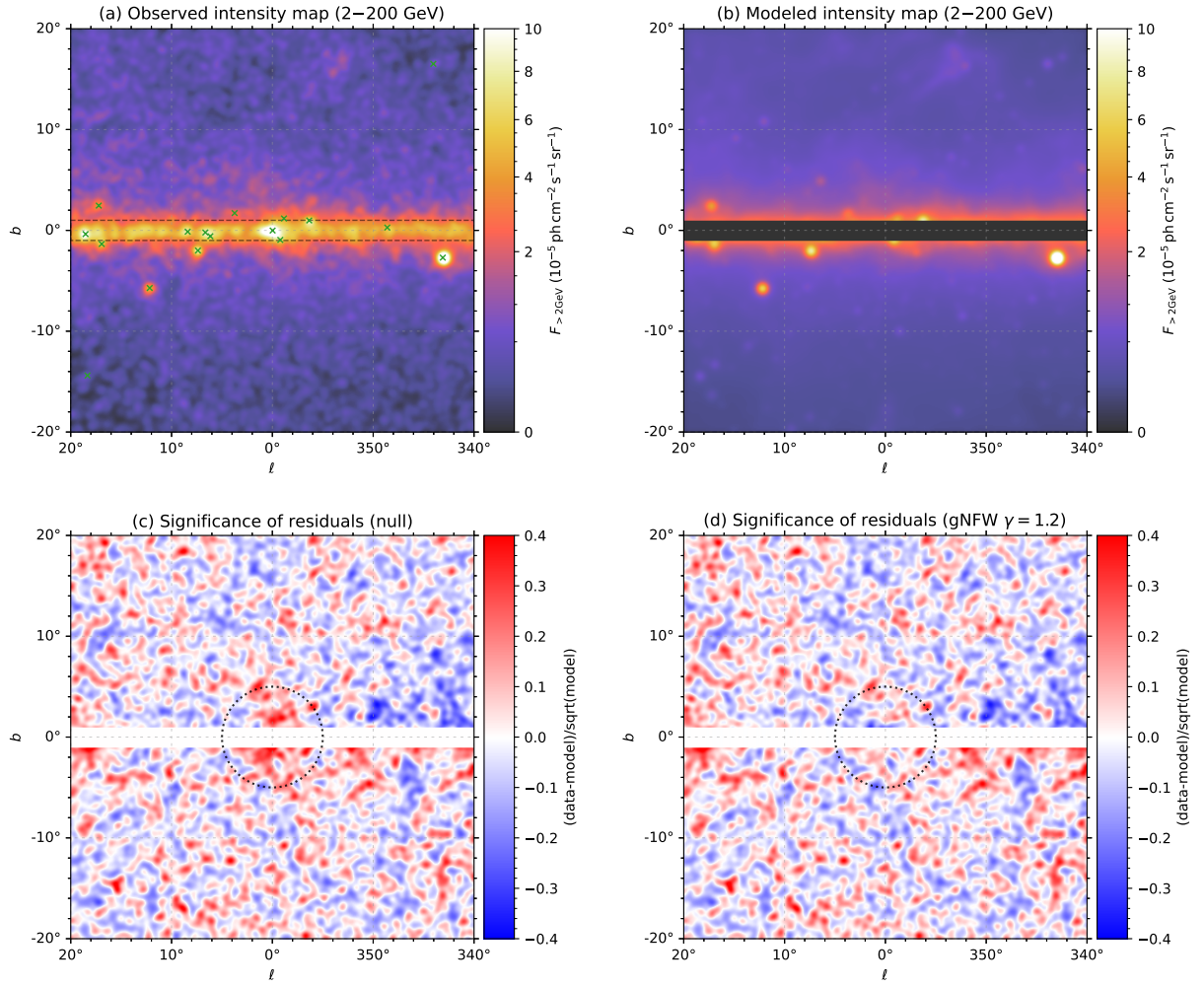


Figure 6. Upper panels: integrated intensity map of the ROI from (a) the observations and (b) the best-fit null model. The green crosses represent the point sources in the DAMPE catalog. The black dashed line encloses the Galactic plane region ($|b| < 1^\circ$) that is masked in the likelihood analysis. Lower panels: the significance of the residual maps for the model (c) without and (d) with the GC excess model. The black dotted line shows a 5° circle centered at the GC. All the maps are smoothed with a $\sigma = 0.3$ Gaussian kernel.

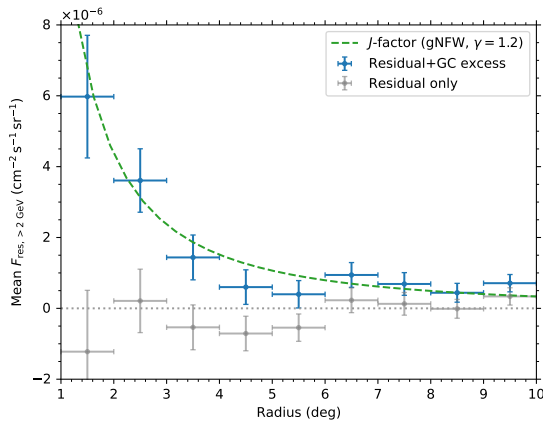


Figure 7. The average residual flux in annuli centered at GC. The gray points show the residual flux given the alternative model, whereas the blue points show the flux that includes both the residual and GC excess. The green dashed line represents the flux of the best-fit J -factor model.

high-energy tail or is merely a statistical fluctuation. If we only account for the first five energy bins (2–20 GeV), which cover the main spectral component of the GC excess, the TS value decreases to 73.8, but the significance increases to 7.7σ .

4.2. Morphological Study

The morphology of the GC excess can help unveil its nature, which is extensively studied in Fermi-LAT observations (O. Macias et al. 2018, 2019; R. Bartels et al. 2018; M. Di Mauro 2021; S. D. McDermott et al. 2023; D. Song et al. 2024; Y.-M. Zhong & I. Cholis 2024; E. D. Ramirez et al. 2025). If the excess originated from DM annihilation, its spatial shape would be brighter in the center and more spherical than an MSP origin. In this subsection, we will compare the excess with the gNFW profile and then test various bulge templates. We adopt the same backgrounds as used in the baseline model in the analysis since it is among the best-fitted models to the data.

We change the inner density slope of the gNFW profile γ , calculate the J -factor template, and repeat the bin-by-bin analysis in the energy range from 2 to 20 GeV. The upper panel of Figure 9 demonstrates the difference of the likelihood value $-2\Delta \ln(\mathcal{L}) \equiv -2 \ln[\mathcal{L}(\gamma)/\mathcal{L}(\gamma_{\text{best}})]$ versus the inner slope γ , where $\mathcal{L}$ is the product of the likelihood values for the lowest five energy bins, and γ_{best} is the optimal density slope. The orange band in the figure shows the 1σ statistical uncertainty of γ . Given the baseline background model, the slope is found to be $\gamma = 1.23 \pm 0.08$ with DAMPE data. Such

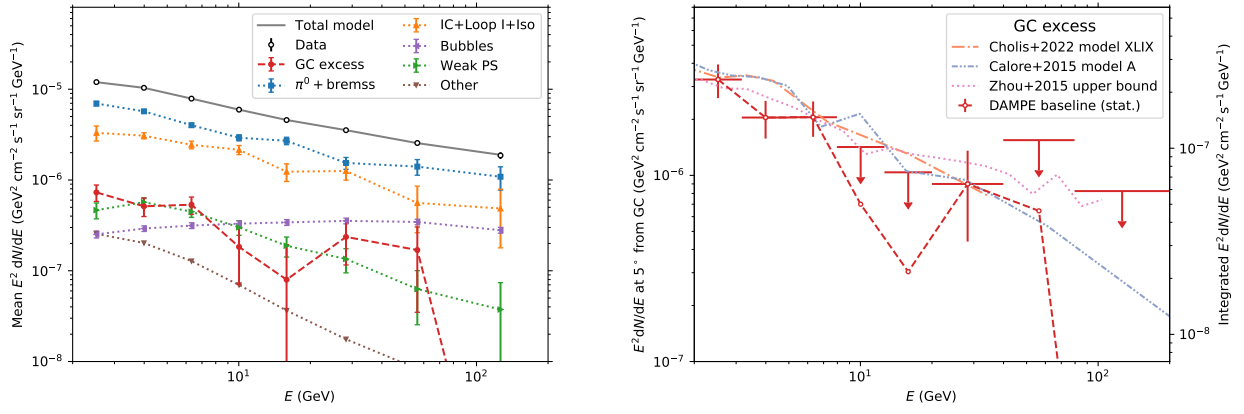


Figure 8. Left panel: the mean flux of the observed data and the components in the baseline model within the ROI. The red points show the flux of the GC excess. The masked regions are excluded from the calculation. Right panel: the baseline SED of the GC excess observed by DAMPE (red points). The left y-axis represents the SED at an angle of 5° from the GC, whereas the right y-axis is the flux integrated over the circle $<10^\circ$ from the GC excluding 2° of the Galactic plane. 95% upper limits are shown when the TS values are below four. The red dashed line connects the best-fit flux points. The GC excess follows the J -factor template given the gNFW profile with $\gamma = 1.2$. Some of the Fermi-LAT observations are also shown with lines.

Table 4

The Spectral Energy Distribution of the Galactic Center Excess at an Angle of 5° from the Galactic Center Given the Baseline Background Model, as in the Right Panel of Figure 8

E Range (GeV)	$E^2 dN/dE$ ($10^{-6} \text{ GeV cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$)	TS Value
2.00–3.17	3.27 ± 0.67	25.3
3.17–5.02	2.05 ± 0.47	21.0
5.02–7.96	2.05 ± 0.44	24.2
7.96–12.6	<1.42	2.9
12.6–20.0	<1.04	0.5
20.0–39.9	0.90 ± 0.46	4.5
39.9–79.6	<1.55	1.8
79.6–200	<0.82	0.0

Note. The error bars only show the statistical uncertainties. 95% upper limits are given when the TS values are below four. The third column shows the TS values in the energy windows.

a density steeper than the regular NFW profile ($\gamma = 1.0$) can be reasonable for the MW due to the contraction of DM halo alongside the accretion of baryons at the GC (M. Cautun et al. 2020). Our result also agrees with the Fermi-LAT observations very well. Shown with the hatched bands are the 1σ errors from T. Daylan et al. (2016) and F. Calore et al. (2015b). Recent analyses (M. Di Mauro 2021; I. Cholis et al. 2022) confirm the preference for the contracted NFW profile with $\gamma \approx 1.2$ – 1.3 .

The ellipticity of the GC excess is also analyzed. Following I. Cholis et al. (2022), we deform the opening angle from the GC ψ by the ellipticity parameter ε such that $\cos(\psi) = \cos(b)\cos(\ell/\varepsilon)$. The spherical symmetric gNFW model corresponds to $\varepsilon = 1$. The template is elongated along the Galactic plane for $\varepsilon > 1$, whereas it is perpendicular to the plane for $\varepsilon < 1$. The inner density slope is kept to $\gamma = 1.2$. The middle panel of Figure 9 demonstrates the likelihood versus the ellipticity. Our analysis slightly favors a spherical shape of the excess given the best-fit ellipticity of $\varepsilon = 1.0^{+0.3}_{-0.2}$, which is consistent with the baseline model of I. Cholis et al. (2022, purple dashed line) within the 2σ uncertainty. A recent simulation shows the DM density profile of MW-like galaxies is flattened along the Galactic plane (M. M. Muru et al. 2025). In their simulations, the minor-to-major axis ratios for the

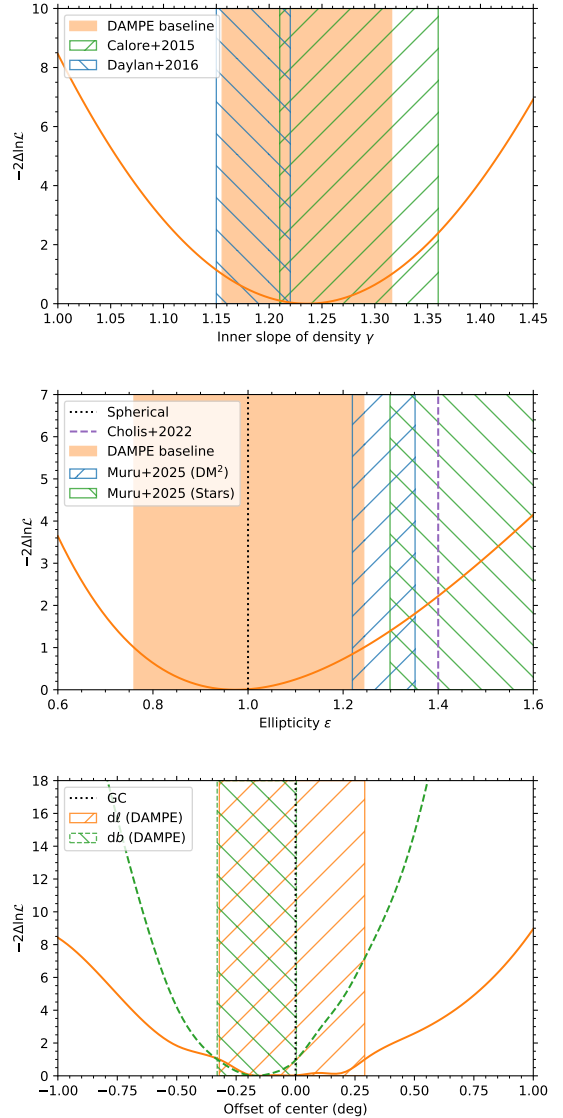


Figure 9. The change of log-likelihood value as a function of the steepness (upper), the ellipticity (middle), and the center position (lower) of the DM density profile derived with the 2–20 GeV data assuming the baseline background model. In the lower panel, the orange solid and green dashed lines show the likelihood variation vs. Galactic longitude and latitude, respectively.

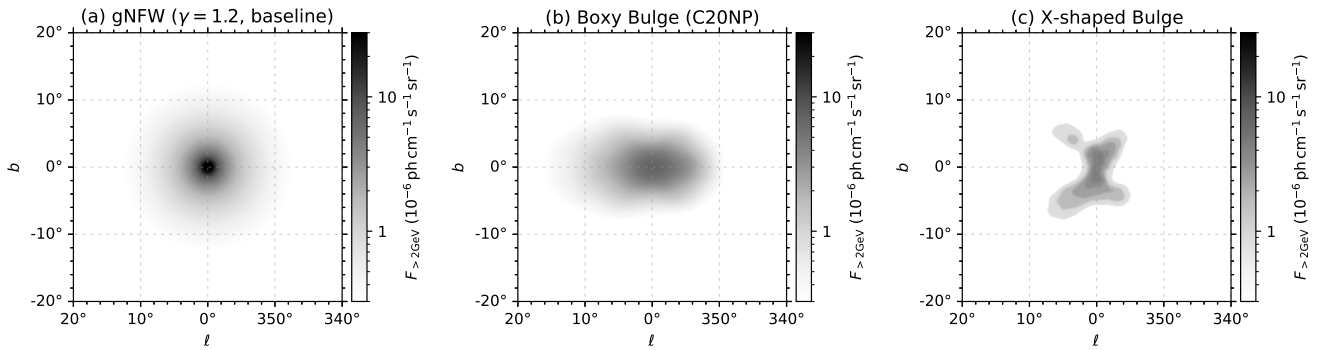


Figure 10. Several spatial models for the GC excess: (a) the J -factor map given the gNFW profile with an inner slope of $\gamma = 1.2$, (b) a nonparametric bulge model (B. Coleman et al. 2020), and (c) an X-shaped bulge map (O. Macias et al. 2018; S. D. McDermott et al. 2023). The intensity of the each component is based on the best-fit model given the GC excess template. We do not convolve the maps with the PSF.

J -factor range from 0.74 to 0.82 in the inner $\sim 5^\circ$ region, which corresponds to $\varepsilon = 1.22\text{--}1.35$ as shown with the blue hatch in the figure. On the other hand, the morphology is slightly more elongated with axis ratios of 0.54–0.77 ($\varepsilon = 1.30\text{--}1.85$). Our baseline result is compatible with the ellipticity of the J -factor in the simulation within the 1σ uncertainty.

The central position of the DM template is also tested. We shift the center of the J -factor template along the Galactic plane and perpendicular to the plane by an angle, conduct the same bin-by-bin analysis between 2 and 20 GeV, and calculate the change of likelihood value with the angle. In this analysis, we assume the spherical gNFW density profile with $\gamma = 1.2$. The result is demonstrated in the lower panel of Figure 9, where the solid orange and dashed green lines are for the models with the longitude and latitude of the center changed, respectively. The optimal longitude of the center is $d\ell = -0.1^{+0.4}_{-0.2}$ given a latitude of $db = 0^\circ$, while the optimal latitude is $db = -0.2 \pm 0.2$ given $d\ell = 0^\circ$. The center of the excess is slightly offset from the dynamic center of the GC, located at its southwest. A similar offset with an angle of ~ 0.1 was reported in Fermi-LAT observations as well (M. Di Mauro 2021; M. Ackermann et al. 2017). Nevertheless, the offset is not significant in the DAMPE data.

Finally, we substitute the J -factor templates with the bulge models, which are considered tracers of MSPs. We adopt the bulge–bar model (model S) developed by H. T. Freudenreich (1998, denoted F98S), the X-shaped bulge model (O. Macias et al. 2018) from S. D. McDermott et al. (2023),³⁵ and the nonparameterized bulge model from B. Coleman et al. (2020, denoted C20NP).³⁶ Some of the bulge models are shown in Figure 10. We use these bulge models as a template of the excess and repeat the bin-by-bin analyses in the energy range of 2–200 GeV. The resultant TS values are given in Table 5. Since they share the same null model, a template with a larger TS value also has a larger likelihood value and thereby fits the data better. The C20NP bulge model is the best among the three bulge models, showing a TS value of 77.8, but it is worse than the baseline J -factor map produced from the gNFW profile. The preference is still valid if we restrict the energy range to 2–20 GeV. The result is not surprising since the spherical shape of the GC excess is slightly preferred as shown previously.

³⁵ <https://github.com/samueldmcdermott/gcepy>

³⁶ https://github.com/chrisgordon1/galactic_bulge_templates

Table 5
Comparison of the Galactic Center Excess Models

Excess Model	TS _{2–200}	TS _{2–20}
gNFW ($\gamma = 1.2$)	80.1	73.8
X-shaped bulge	66.4	62.8
Boxy bulge (F98S)	71.0	64.7
Boxy bulge (C20NP)	77.8	70.8

Note. The second (third) column is the TS value of the excess for the data from 2 to 200 GeV (20 GeV). The baseline background model described in Section 2.2 is adopted.

4.3. Systematic Uncertainty

Though the presence of the GC excess is quite robust, its flux and spectrum are significantly influenced by systematic uncertainties as revealed in the data analysis of Fermi-LAT (e.g., F. Calore et al. 2015b; B. Zhou et al. 2015; M. Ackermann et al. 2017; I. Cholis et al. 2022). In this subsection, we will discuss the impact of the mask, the Fermi bubbles template, the source model, and the GDE templates on the observation of the GC excess with DAMPE. We still adopt the J -factor map with a gNFW density slope of $\gamma = 1.2$ as the template of the GC excess. The main results are given in Figure 11 and Table 6.

First, we test three types of masks in order to understand the impact of the Galactic plane, the excess in the residual map, and the events from point sources distributed outside the mask. In the first case, we extend the Galactic plane mask to $|b| < 2^\circ$. In the second case, we exclude the region with $340^\circ < \ell < 355^\circ$ and $|b| < 10^\circ$ where some excess and deficit exist in the residual map (see Figure 6(d)). Such structures are also presented in Fermi-LAT data (e.g., F. Calore et al. 2015b). In the last case, we adopt a larger point-source mask with a radius of 1° . The resultant spectra of the GC excess are shown in Figure 11(a) along with the TS values listed in the label. In all three cases, the significance decreases a little. For cases 1 and 3, the decrease is likely caused by the removal of the signal region since their SEDs are close to or even higher than the baseline. However, for case 2, the spectrum is lower than the baseline, so the low TS value of ~ 48.3 is caused by absorption of the GC excess by the background. Interestingly, the SED point at ~ 16 GeV increases a bit in this case, so the weak disagreement with the Fermi-LAT data may have a systematic origin.

The low-latitude Fermi bubbles also affect the spectrum of the GC excess. However, due to the low statistics and complex

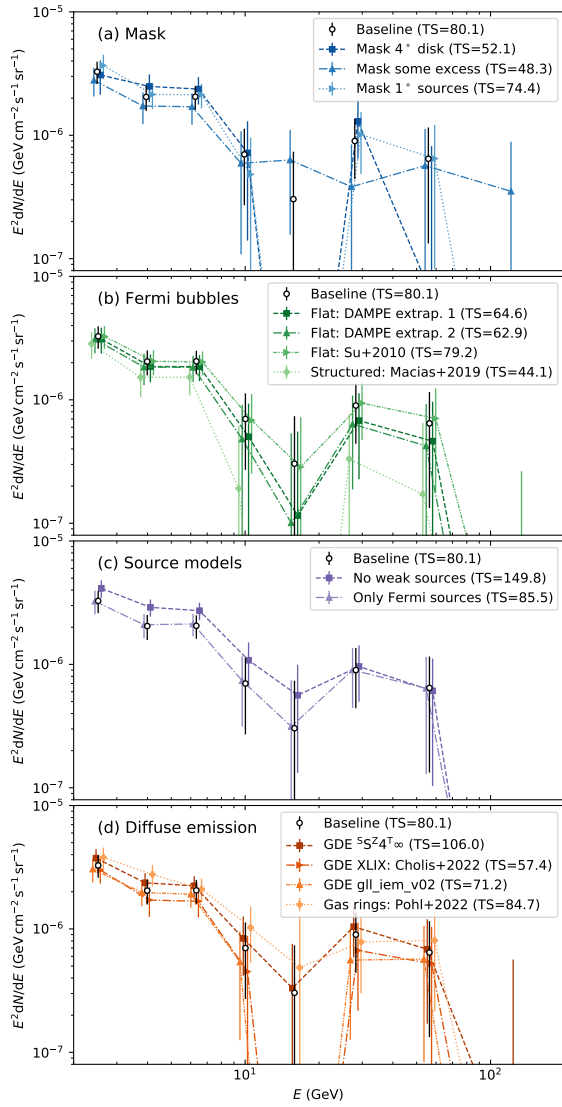


Figure 11. The systematic uncertainty of the GC excess caused by the analysis procedures and the background models, including (a) the mask, (b) the Fermi bubbles template, (c) the point and extended sources, and (d) the GDE models. The points show the SEDs and the statistical uncertainties measured at 5° from the GC. The TS values in the labels are for the 2–200 GeV data. The central energies are slightly offset to avoid overlapping.

background, it is hard for DAMPE to make such a template. We test four bubbles templates with low-latitude structure for the systematic uncertainty (Figure 12). Two templates are based on the Fermi-LAT observation: one is the flat template extracted from the residual map in M. Su et al. (2010; Figure 12(c)), and the other one is a structured map (O. Macias et al. 2019) after applying the Laplace in-painting algorithm to the hard component in the residuals (M. Ackermann et al. 2017; Figure 12(d)). Two other templates are extrapolated from the baseline Fermi bubbles templates extracted in Section 3.1. In the first case (denoted as *extrap.1*), we connect the base of the bubbles with two lines to make a intersection point, and define the low-latitude bubbles with the northern and southern parts among the four regions separated by the lines (see Figure 12(a)). In the second case (*extrap.2*), a low-latitude template is extracted based on the shape of the in-painted Fermi bubbles template. But different from the latter one, we exclude some excess along the

Table 6
Comparison of the Galactic Center Excess in Various Background Models

Background	$\ln(\mathcal{L}_{\text{gNFW}})$	TS_{gNFW}	TS_{C20NP}
Baseline	−129,068.8	80.1	77.8
Bubbles <i>extrap.1</i>	−129,066.3	64.6	61.0
Bubbles <i>extrap.2</i>	−129,065.1	62.9	57.6
Bubbles (M. Su et al. 2010)	−129,085.8	79.2	69.2
Bubbles (O. Macias et al. 2019)	−129,070.0	44.1	37.3
No weak sources	−129,248.1	149.8	153.5
Only Fermi sources	−129,083.6	85.5	78.8
GDE $\text{S}^{\text{Z}4^{\text{T}}}_{\infty}$	−129,069.5	106.0	101.1
GDE XLIX	−129,130.3	57.4	69.4
GDE <i>gll_iem_v02</i>	−129,225.3	71.2	103.6
GDE (M. Pohl et al. 2022a)	−129,155.6	84.7	120.1

Note. The second column shows the log-likelihood values with the gNFW ($\gamma = 1.2$) model of GC excess using the 2–200 GeV data. The third and fourth columns show the TS values of the excess given the gNFW and boxy bulge (C20NP) templates, respectively. The baseline mask is adopted throughout the analyses.

disk and assume the template is uniform (Figure 12(b)). The baseline Fermi bubbles template is substituted, and the same fittings are performed. The results are shown in Figure 11(b). The extrapolated bubbles templates fit the data slightly better than the baseline template. For the *extrap.1* bubbles template, the log-likelihood increase is 2.4 and the TS value for the GC excess is 64.6 (6.5σ); whereas for the *extrap.2* template, the log-likelihood increase is 3.6 and the TS value is 62.9 (6.4σ). The spectra of the GC excess are also very similar in the two cases. On the other hand, the two bubbles templates from Fermi-LAT perform worse than the baseline. The log-likelihood decreases are 3.3 and 18.1 for the structured map (O. Macias et al. 2019) and flat map (M. Su et al. 2010), respectively. The structured bubbles template is also peaked at the GC, so it may absorb emission from the GC excess, causing a smaller TS value of 44.1 (5.0σ).

Furthermore, the effect of point and extended sources is checked. We either remove the weak source template from the model or replace all the point and extended sources with the ones defined in the Fermi-LAT 14 yr catalog (J. Ballet et al. 2023) and then fit them as a whole. As shown in Table 6, the fitting deteriorates if the weak sources are not accounted for. However, using the Fermi-LAT source catalog is statistically slightly better than the baseline model according to the Akaike information criterion (H. Akaike 1974), considering the baseline model has $15 \times 2 = 30$ more free spectral parameters for the sources. The corresponding GC excess spectra are presented in Figure 11(c). The low-energy part of the spectra can be affected by weak sources, as illustrated in the first case. This is because the weak sources are concentrated at the center and can be absorbed by the GC excess if this component is not accounted for. However, the sources that are even unresolved by Fermi-LAT cannot be verified here. We also find that altering the source catalog makes little impact on the SED.

Finally, the GDE models are inspected. We show four models as representatives. The first one is from Fermi-LAT 1SC (F. Acero et al. 2016a), labeled as $\text{S}^{\text{Z}4^{\text{T}}}_{\infty}$, which assumes the CR source distribution traced by supernova remnants, a height of the propagation halo of 4 kpc, and optically thin H I gas. The second one is the Galprop-based GDE model XLIX developed by I. Cholis et al. (2022). They fitted injection and propagation parameters to AMS-02 CR

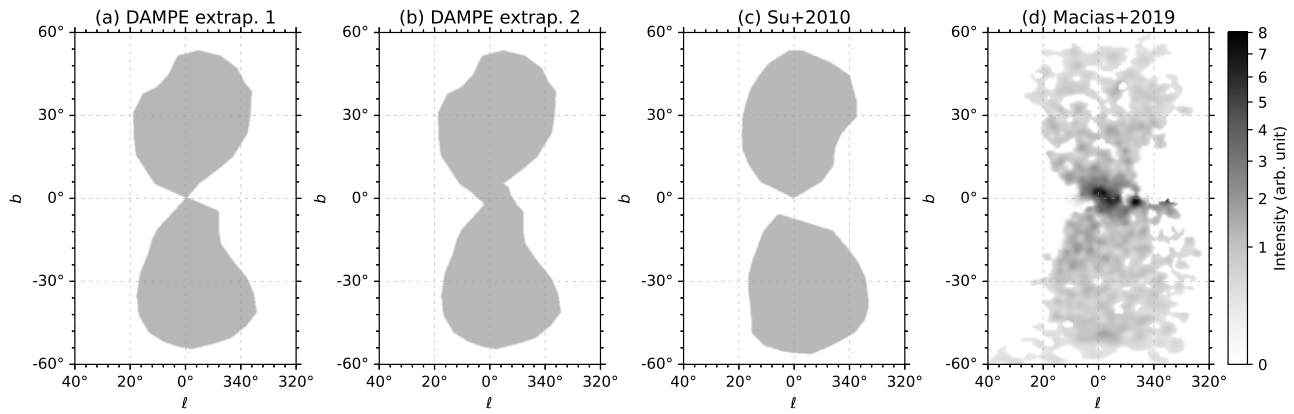


Figure 12. Alternative bubbles templates with a low-latitude component for evaluating the systematic uncertainty of the GC excess: (a)–(b) the extrapolated maps from the baseline template and (c)–(d) the bubbles derived from Fermi-LAT observations.

data (e.g., M. Aguilar et al. 2018) and generated 80 alternative CR parameter sets and γ -ray templates. They also accounted for different tracers of the CR electron and proton sources. The model XLIX was reported as the best fit to the Fermi-LAT data among all the alternatives. We take the parameter set (I. Cholis et al. 2021), calculate the GDE templates using `Galprop v.54.1.984` (A. W. Strong & I. V. Moskalenko 1998), and combine the rings together to make the three components the same as the baseline model. The third one is the data-driven model `gll_iem_v02` from Fermi-LAT,³⁷ which is often adopted in analysis of the GC excess (e.g., T. Daylan et al. 2016). It consists of hadronic and bremsstrahlung emission traced by gas column densities in six Galactocentric rings and IC emission from `Galprop`,³⁸ so we replace the three GDE components with this model. The final model is taken from M. Pohl et al. (2022a, 2022b). They provide new gas column density maps split into four rings and IC templates considering both CR sources along the spiral arms (G. Jóhannesson et al. 2018) and the 3D distribution of the ISRF (T. A. Porter et al. 2017). We keep the four gas rings for H I and H₂, assign PL scales to them, and tie the spectral indices of the neutral and molecular gas templates in each ring together. For the IC components, we add all six rings to make a single IC template. The positive dust residuals are also included in the fitting as an independent component. All the fitted spectra are shown in Figure 11(d). The latter three GDE models present worse fitting than the baseline model, with the log-likelihood differences larger than 60, while the first model fits the data similarly well to the baseline model. We check the residual maps of the three models and find slightly more positive residuals along the southern Galactic plane. In all these cases, the GC excess shows significant emission with the lowest TS value of 57.4 for the GDE model XLIX, corresponding to a significance of 6.0σ . The low-energy component below ~ 10 GeV and high-energy tail around 40 GeV persist, while the dip at ~ 16 GeV is prone to the choice of the GDE model.

We also repeat the systematic analysis assuming the C20NP bulge template for the GC excess (Figure 10(b)). The TS values are presented in the fourth column of Table 6. The GDE is a major systematic factor for morphological studies of the excess. If the baseline GDE model ($^{\text{SLZ}}10^{\text{T}}150$) is adopted, the bulge template is generally less favorable than the gNFW

template. The preference persists given the GDE model $^{\text{S}}\text{S}^{\text{Z}}4^{\text{T}}\infty$, whose log likelihood ranks second among the five GDE models. But the C20NP bulge model is preferred in the remaining three models. For the last two GDE models in the table, the bulge is much better than in the gNFW parameterization. This is because more positive residuals exist along the Galactic plane compared to the others, which are absorbed by the box bulge template. We further test the X-shape bulge (Figure 10(c)). The TS values of the bulge are 78.8, 56.1, 64.3, and 69.3 for the four GDE models, respectively. The X-shaped bulge is worse than the gNFW template in these cases.

4.4. Implications on the Dark Matter Parameters

If the GC excess originates from DM annihilation, we can set constraints on the DM parameters. The spectrum of the prompt emission emitted by annihilating DM particles can be calculated with

$$\frac{dN}{dE} = \frac{1}{4\pi} \frac{\langle\sigma v\rangle_f}{2m_\chi^2} \frac{dN_f}{dE} \times J, \quad (5)$$

where m_χ is the mass of the DM particle, and $\langle\sigma v\rangle_f$ is the velocity-averaged annihilation cross section in channel f . They are the two free parameters in the fittings. dN_f/dE is the DM prompt γ -ray spectrum per annihilation, which is interpolated from the tabulated data in PPPC4 (M. Cirelli et al. 2011). J is the J -factor. Since the spectrum is normalized at 5° from the GC, the J -factor should also be at $\psi = 5^\circ$. In this work, we focus on two annihilation channels: $\chi\chi \rightarrow b\bar{b}$ and $\chi\chi \rightarrow \tau^+\tau^-$. We fit the DM spectrum to the SED of the GC excess. The likelihood profiles in all eight energy bins are adopted, similar to Section 3.2.

Figure 13 exhibits the DM annihilation models best fitted to the baseline spectrum along with the 1σ statistical uncertainty bands. Both DM models can well describe the emission in the low-energy range. For the annihilation channel to b quarks, the DM mass needs to be $m_\chi = 49_{-12}^{+16}$ GeV and the cross section is $\langle\sigma v\rangle = 1.9_{-0.3}^{+0.4} \times 10^{-26} \text{ cm}^3 \text{ s}^{-1}$. For the channel to the leptons $\tau^+\tau^-$, the optimal DM mass is $m_\chi = 13.0_{-1.8}^{+1.9}$ GeV and the annihilation cross section is $\langle\sigma v\rangle = (6.0 \pm 0.9) \times 10^{-27} \text{ cm}^3 \text{ s}^{-1}$. The 1σ and 2σ contours, defined with the log-likelihood differences relative to the optimal model ($-2\Delta\ln(\mathcal{L})$) of 2.30 and 6.18, are presented with the solid and dashed red curves in Figure 14, respectively. We also show the systematic uncertainties of the DM parameters for

³⁷ <https://fermi.gsfc.nasa.gov/ssc/data/access/lat/BackgroundModels.html>

³⁸ https://fermi.gsfc.nasa.gov/ssc/data/access/lat/ring_for_FSSC_final4.pdf

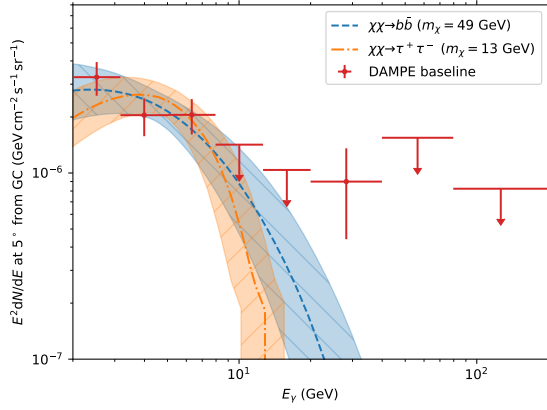


Figure 13. The SED of the GC excess (red points) for the baseline background and the best-fit DM annihilation models. The blue dashed and orange dotted-dashed lines correspond to the DM annihilation channels of $\chi\chi \rightarrow b\bar{b}$ and $\chi\chi \rightarrow \tau^+\tau^-$, respectively. The colored bands show the 1σ statistical uncertainties of the DM models.

various background models (Section 4.3) with the error bars, which are as large as the statistical uncertainties. The 2σ regions from the Fermi-LAT observations of the GC excess conducted by F. Calore et al. (2015b, dotted-dashed line) and I. Cholis et al. (2022, dotted-dotted-dashed line) are given in the figure. In both channels, our parameter spaces are consistent with the previous works based on the Fermi-LAT data. Drawn with the blue dotted line in the figures are the 95% confidence level upper limits of the cross sections from dSphs observed by Fermi-LAT (A. McDaniel et al. 2024). Some of the DM parameter spaces for the two channels are excluded by the dwarf galaxies.

5. Conclusion

The GC region, which hosts an SMBH and a large amount of DM as well as CRs, is the most extreme region in the MW and has attracted wide interest in the high-energy astrophysics community. Among the diffuse γ -ray sources detected in the GC region, the large-scale bubbles and the GC excess in the GeV band detected first by Fermi-LAT have attracted wide attention. In particular, a DM origin of the GC excess emerging in the GeV band has been widely speculated in the literature and a multimessenger “counterpart” may have been identified in AMS-02 antiproton data.

Besides Fermi-LAT, DAMPE is the only other GeV γ -ray detector currently in operation in space that has collected a large amount of data. With photon events above 2 GeV observed from 2016 January to 2024 June, the Fermi bubbles and the GC excess are detected with significances of $\sim 26\sigma$ and $\sim 7\sigma$, respectively. Both sources are robust against systematic uncertainties. Their morphology and spectra are well consistent with those measured with Fermi-LAT. In particular, the GeV excess component can be interpreted by DM annihilation with a mass of ~ 50 GeV and a velocity-averaged cross section of $\sim 10^{-26} \text{ cm}^3 \text{ s}^{-1}$ for the $\chi\chi \rightarrow b\bar{b}$ channel, also in agreement with previous findings made with the Fermi-LAT data.

A new space mission, the Very Large Area γ -ray Space Telescope (VLAST), has been proposed. The VLAST mission (Y. Z. Fan et al. 2022; Q. Wan et al. 2023; X. Pan et al. 2024; Y. Zhang et al. 2025; Z. Yang et al. 2026), is distinguished by its extremely wide energy range (from ~ 1 MeV to at least 10 TeV) and a very large acceptance (with a peak of $\sim 12 \text{ m}^2 \text{ sr}$). If successfully launched, the nature of the Fermi

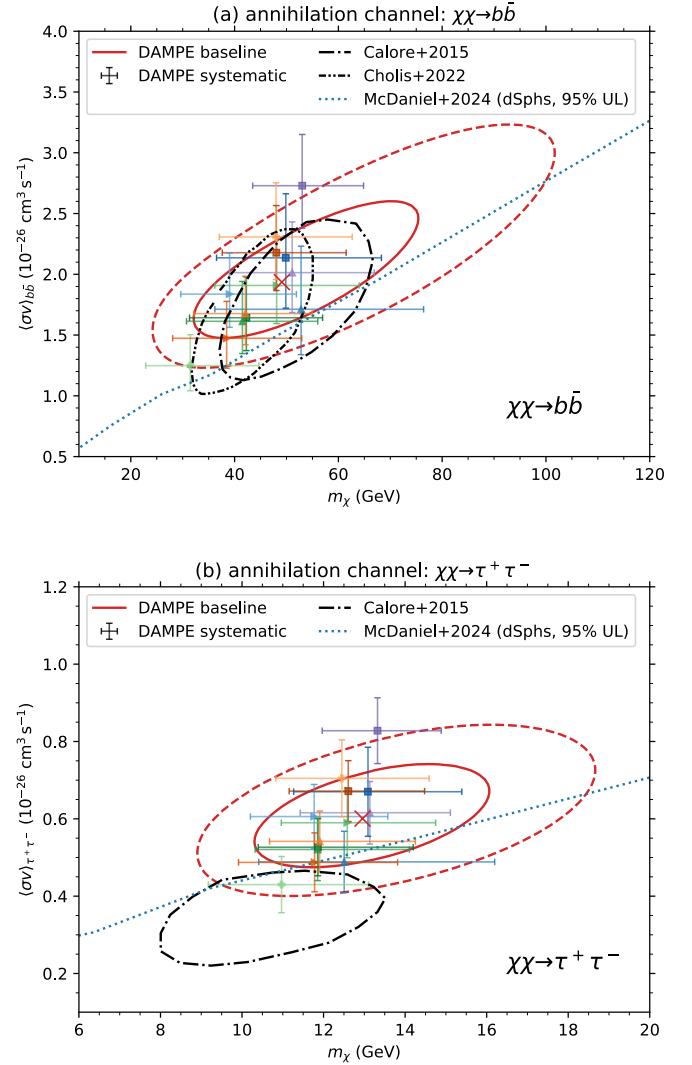


Figure 14. The preferred DM parameters for the annihilation channels of (a) $\chi\chi \rightarrow b\bar{b}$ and (b) $\chi\chi \rightarrow \tau^+\tau^-$. The red crosses are the best-fit points. The solid and dashed red contours are the 1σ and 2σ regions, respectively. The error bars show the DM parameters and the statistical uncertainties, given various background models with the same colors as those in Figure 11. The dotted-dashed and dotted lines are the 2σ contours from the Fermi-LAT observations of the GC excess. The blue dotted lines are the 95% confidence level upper limits from dSphs observed by Fermi-LAT.

bubbles as well as the GeV excess in the GC will be further revealed.

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Software: NumPy (C. R. Harris et al. 2020), SciPy (P. Virtanen et al. 2020), Matplotlib (J. D. Hunter 2007), Astropy (A. M. Price-Whelan et al. 2022), healpy (K. M. Górski et al. 2005; A. Zonca et al. 2019), iminuit (H. Dembinski et al. 2020; F. James & M. Roos 1975), Galprop (A. W. Strong & I. V. Moskalenko 1998; A. W. Strong et al. 2000), DmpST (K.-K. Duan et al. 2019).

Author Contributions

This work is the result of the contributions and efforts of all the participating institutes. All authors have reviewed, discussed, and commented on the results and on the manuscript. In line with the collaboration policy, the authors are listed alphabetically.

Appendix A

Maps of the γ -Ray Components in the Baseline Model

In Section 2.2 and Table 2, we introduce the γ -ray-emitting components of the baseline model. To better understand the structures of the components, we

show their intensity maps within the ROIs of the two targets in this section.

Figure 15 presents the intensity maps and mask adopted in the analyses of the Fermi bubbles. The first two subfigures show (a) a map of hadronic and bremsstrahlung emission and (b) a map of IC emission from the Fermi-LAT supernova catalog (F. Acero et al. 2016a). The third plot shows the template of Fermi bubbles derived with DAMPE data, while the fourth presents the geometric template of Loop I (M. Wolleben 2007). To better visualize the structures in these components, the maps are not convolved with the PSF. The final intensity map is for the bright point sources, whose parameters are from the 8.7 yr DAMPE source catalog (DAMPE Collaboration 2026, in preparation). The intensities of all the components are calculated based on the optimal spectral parameters of the alternative model, given the baseline background emission model.

Figure 16 presents the intensity maps and mask adopted in the analyses of the GC excess. The first four maps are for (a) hadronic and bremsstrahlung emission, (b) IC emission, (c) the Fermi bubbles, and (d) bright point sources, respectively. The fifth map shows the weak sources that are included in the Fermi-LAT 4FGL-DR4 source catalog but not significantly detected by DAMPE. The intensities are based on the best-fit model containing the gNFW template with a slope of $\gamma = 1.2$.

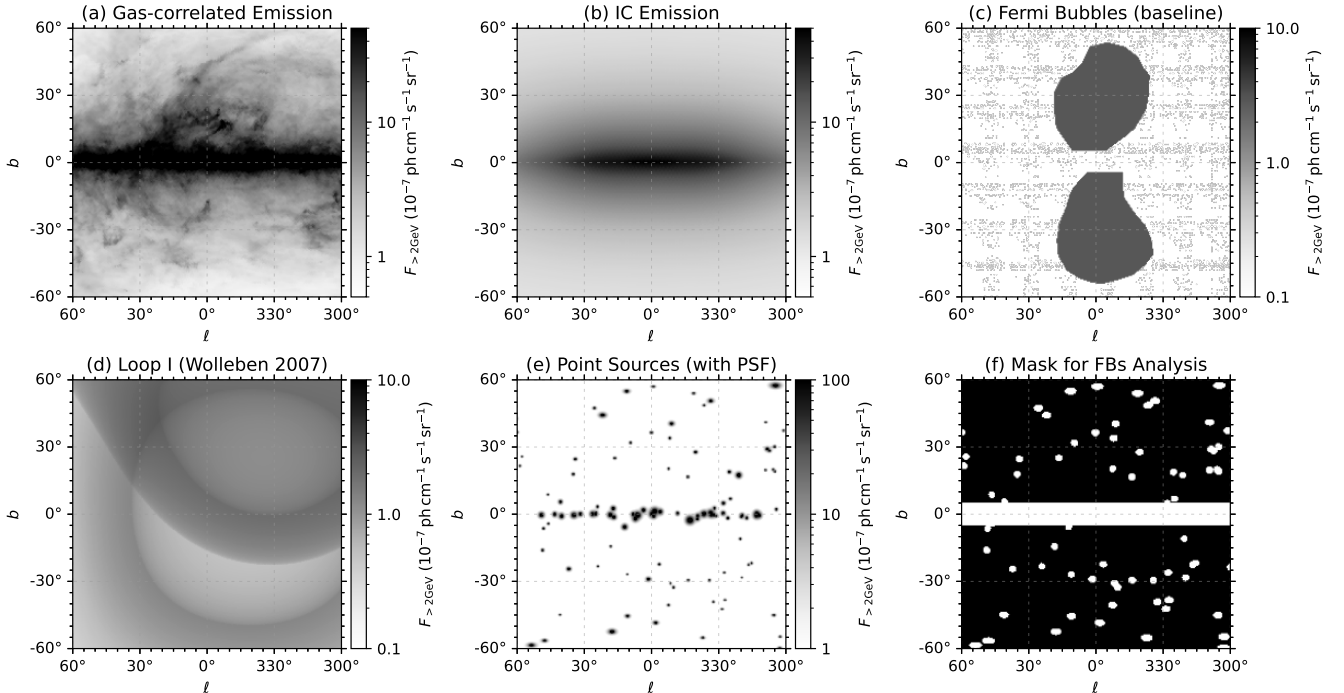


Figure 15. (a)–(e) Maps of the components and (f) mask in the analyses of the Fermi bubbles. The maps include (a) hadronic and bremsstrahlung emission associated with the gas, (b) IC emission, (c) flat template of Fermi bubbles, (d) geometric template of Loop I, and (e) bright point sources. We do not convolve the first four maps with the PSF. The white regions in the mask are excluded from the data analysis. Both the maps and mask are converted from the HEALPix projection.

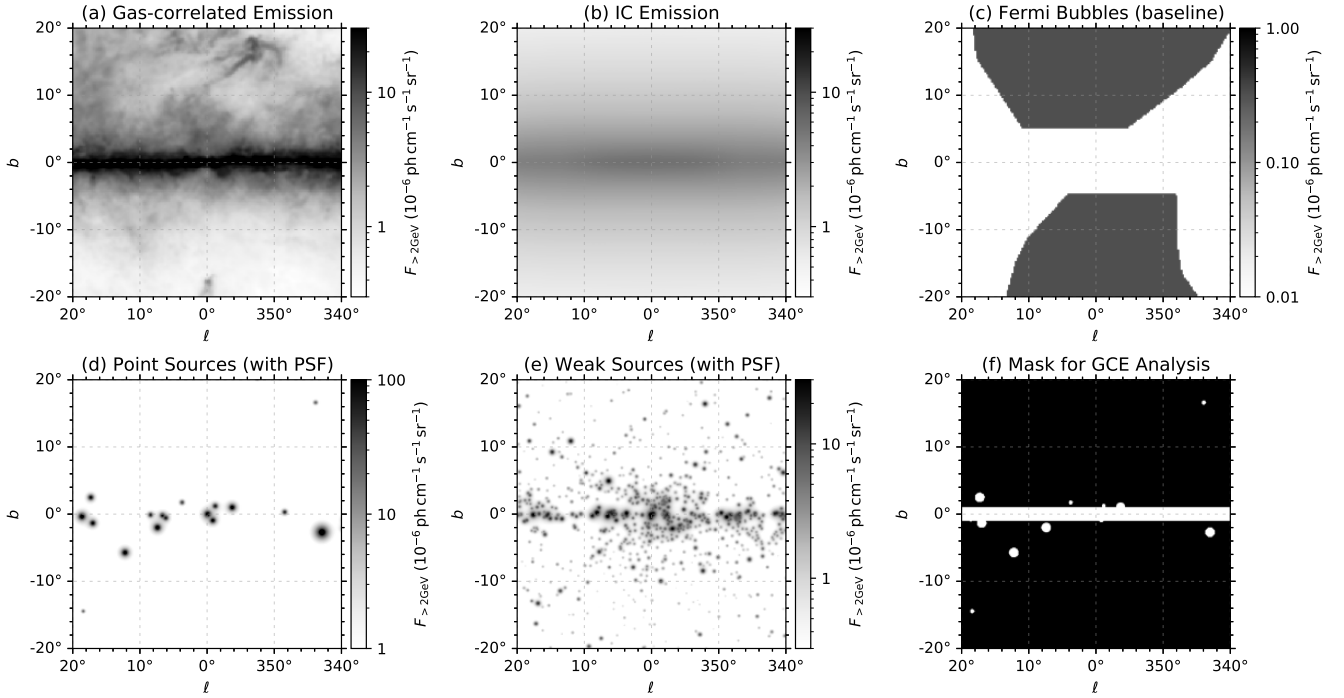


Figure 16. (a)–(e) Maps of the components and (f) mask in the analyses of the GC excess. The first four components are the same as those in the analyses of the bubbles. The color bars are adjusted for better visualization. The fifth map is for the weak sources detected by Fermi but not by DAMPE.

Appendix B

Global Spectra of the γ -Ray Components in the Baseline Model

We use the best-fit global models to calculate residual maps in the analyses of the Fermi bubbles (Section 3) and GC excess (Section 4). In this section, we present the mean fluxes of the

components within the ROI for the best-fit global models. The left and right panels of Figure 17 show the component fluxes from the best-fit global null and alternative models for the Fermi bubbles, whereas Figure 18 is for the GC excess. The data points correspond to the SEDs from the bin-by-bin analyses.

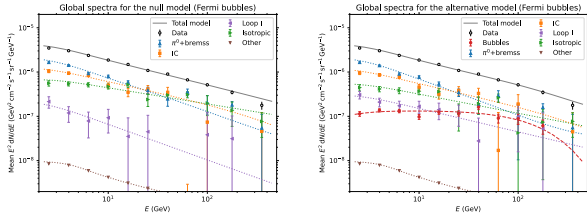


Figure 17. The mean fluxes of the components within the ROI for the best-fit global null (left) and alternative (right) models of the Fermi bubbles analysis. The data points show the best-fit SEDs from the bin-by-bin analysis.

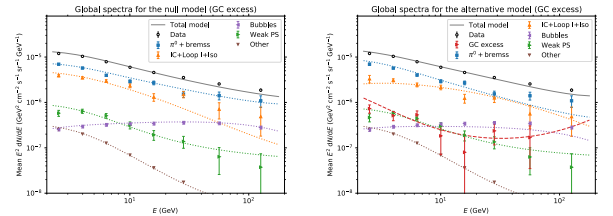


Figure 18. The mean fluxes of the components within the ROI for the best-fit global null (left) and alternative (right) models of the GC excess analysis.

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